

Notes: Gompers and Metrick (QJE, 2001)

Institutional Investors and Equity Prices

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1 One-sentence summary

Gompers and Metrick (2001) argue that the rapid rise of large institutional investors from 1980 to 1996 changed the representative investor's demand for stock characteristics, raising demand for large and liquid stocks and helping explain the disappearance of the historical small-stock premium.

2 Big picture

The paper studies how the composition of equity ownership can affect prices. Traditional asset-pricing models often treat demand curves for stocks as very elastic: if investors demand more of one stock or class of stocks, arbitrageurs should absorb the order flow without much price pressure. Gompers and Metrick study a setting where this view may be too strong. They show that large institutions became much more important owners of U.S. equities over the 1980s and 1990s, and that these institutions systematically preferred particular stock characteristics.

The main fact is simple: large institutions nearly doubled their share of the U.S. equity market, from 26.8 percent in March 1980 to 51.6 percent in December 1996. If institutions and individuals wanted the same stocks, this would not matter much for relative prices. But institutions are not neutral investors. They demand large, liquid stocks and avoid recent winners. As investment discretion shifts from individuals to institutions, aggregate demand shifts toward the characteristics institutions prefer.

The paper then asks whether this demand shift is large enough to matter for equilibrium prices and returns. Its answer is yes. Institutional ownership forecasts future stock returns, especially in quarters when aggregate inflows to institutions are high. In a calibration, the composition shift alone can generate a nearly 50 percent relative price increase of large-company stocks versus small-company stocks under unit-elastic demand.

3 Research question

The paper asks three connected questions.

1. **How did institutional ownership change from 1980 to 1996?** The authors document both the growth of all 13F-reporting institutions and the increasing concentration of holdings among the largest institutions.
2. **Do institutions demand different stock characteristics than individuals?** The paper estimates whether institutional ownership is related to size, liquidity, prudence proxies, book-to-market, and past returns.
3. **Do institutional demand shifts affect equity prices and returns?** The authors test whether institutional ownership predicts returns and calibrate whether the ownership-composition shift can explain the reversal of the small-stock premium.

4 Incremental contribution

4.1 From institutional growth to asset pricing

Earlier research documented the growth of institutional investors, and other work studied institutional portfolio preferences. The incremental step here is to connect those two facts to a broad asset-pricing implication. The paper shows that if institutions have stable preferences and their market share rises sharply, then the representative investor's demand curve shifts in a predictable direction.

This transforms institutional ownership from a descriptive ownership fact into a potential explanation for systematic changes in returns across size groups.

4.2 Combining time-series and cross-sectional demand variation

The paper's identification combines a time-series change and a cross-sectional preference. The time-series change is the growth of institutions. The cross-sectional preference is that institutions prefer large and liquid stocks. The interaction implies rising relative demand for large liquid stocks over time.

This design is stronger than using only aggregate flows or only cross-sectional ownership. It allows the authors to ask whether stocks preferred by the growing investor clientele earned higher returns during the period of clientele growth.

4.3 Direct measurement of institutional portfolios

The authors use SEC Form 13F data, which report the common-stock holdings of institutions with at least \$100 million under discretionary management. This gives a direct measure of institutional ownership rather than an indirect proxy.

The data allow the paper to form an aggregate institutional portfolio and compare it with the portfolio implicitly held by noninstitutional investors. This is why Figure II can show the return path of institutional holdings relative to individual holdings.

4.4 A demand-shock interpretation of the size-premium reversal

The paper's calibration connects the compositional shift in ownership to the disappearance of the small-stock premium. Institutions devote much larger portfolio shares to large stocks than individuals do. When institutions go from controlling about one-quarter to more than one-half of the market, demand for large stocks rises and demand for small stocks falls. With unit-elastic demand, the relative price effect is large enough to account for much of the return advantage of large stocks after 1980.

5 Data and institutional ownership measurement

5.1 Form 13F holdings

The empirical analysis uses quarterly Form 13F holdings from the first quarter of 1980 through the fourth quarter of 1996. Institutions must file if they have at least \$100 million of securities under discretionary management. The paper focuses on common stocks in CRSP, covering NYSE, AMEX, and NASDAQ National Market System stocks.

For each stock-quarter, institutional ownership is

$$IO_{i,t} = \frac{\text{shares of stock } i \text{ held by 13F institutions at } t}{\text{total shares outstanding of stock } i \text{ at } t}.$$

The authors set $IO_{i,t} = 0$ for CRSP stocks not held by any reporting institution.

5.2 Manager types

The 13F data classify managers into five groups: banks, insurance companies, investment companies or mutual funds, investment advisers, and other institutions such as pension funds and university endowments. The categorization is imperfect, but it is useful for documenting the changing composition of institutional holdings.

A central ownership shift occurs within the institutional sector itself. Banks decline in relative importance, while investment advisers and mutual funds grow sharply.

5.3 Market share growth

Large institutions controlled 26.8 percent of the publicly traded equity market in March 1980 and 51.6 percent in December 1996. The growth is also concentrated among the biggest managers: the one hundred largest institutions increased their market share from 19.0 percent to 37.1 percent.

This concentration matters because the paper is not merely about more institutions entering. It is about a large share of investment discretion moving to very large professional managers with similar revealed preferences.

6 Why institutions may demand different stocks

6.1 Agency and monitoring frictions

Institutions often invest as agents for ultimate beneficiaries. Mutual funds, banks, insurance companies, investment advisers, pension funds, and endowments all face different legal, reporting, and reputational constraints than individual investors. Because clients cannot perfectly monitor institutional decisions, institutional incentives can differ from those of the underlying savers.

This agency channel makes it plausible that institutions systematically favor stocks that are easier to justify, easier to trade, or easier to explain to clients.

6.2 Prudence motives

The paper uses “prudence” to describe fiduciary or regulatory incentives to hold safer, more established stocks. Del Guercio’s prudent-man-rule argument suggests that institutions, especially banks, may prefer stocks that are older, pay dividends, are in the S&P 500, and have lower volatility.

The paper uses age, dividend yield, S&P 500 membership, and stock-return volatility as prudence proxies.

6.3 Liquidity and transaction-cost motives

Institutions trade large positions. Large trades are easier and cheaper in liquid stocks. The paper uses size, share price, and turnover as liquidity proxies. If institutional investors value liquidity more than individuals do, institutional ownership should be higher for large, high-price, high-turnover stocks.

This prediction receives the strongest support in the data.

6.4 Historical return characteristics

The paper also examines whether institutions chase or avoid characteristics associated with historical return patterns: size, book-to-market, and momentum. The results show that institutions are not momentum investors. Conditional on size and other characteristics, institutions hold less of stocks with high recent returns.

7 Determinants of institutional ownership

7.1 Cross-sectional regression

For each quarter, the authors estimate

$$IO_{i,t} = \alpha_t + \beta_t X_{i,t} + \varepsilon_{i,t},$$

where $X_{i,t}$ includes size, book-to-market, dividend yield, price, S&P 500 membership, volatility, age, recent returns, and turnover.

The main coefficients are summarized across 68 quarterly cross-sectional regressions.

7.2 Main ownership patterns

The strongest result is institutional demand for liquidity. Size, turnover, and price have positive coefficients in almost every quarter. S&P 500 membership and age are also generally positively related to institutional ownership.

The momentum results are striking: both recent-quarter momentum and prior-nine-month momentum enter negatively. This means institutions are not simply chasing winners. Once size and other variables are controlled, institutions tilt away from high past returns.

7.3 Stable preferences imply shifting aggregate demand

The institutional preference pattern is stable over time. Since institutional market share rises sharply while preferences remain stable, the paper interprets the ownership shift as a movement from the representative individual investor toward the representative institutional investor. This is the foundation for the demand-shock interpretation.

8 Institutional ownership and returns

8.1 Aggregate institutional portfolio

The authors form a value-weighted portfolio of all stocks held by institutions at the beginning of each quarter and compare it with the corresponding portfolio held by noninstitutional investors. One dollar invested in the institutional portfolio in March 1980 grows to \$13.49 by December 1996; one dollar invested in the individual portfolio grows to \$12.10.

The annualized returns are 15.11 percent and 14.44 percent, respectively. The difference is not huge in annual terms, but it compounds meaningfully and motivates the formal return regressions.

8.2 Return forecasting regression

The paper estimates quarterly Fama-MacBeth-style cross-sectional regressions:

$$r_{i,t+1} = \alpha_t + \beta_t X_{i,t} + \gamma_t IO_{i,t} + \varepsilon_{i,t},$$

weighted by market capitalization. The vector $X_{i,t}$ is the same set of characteristics used in the institutional ownership regressions.

The coefficient on institutional ownership is positive and statistically significant. Stocks with high institutional ownership earn higher next-quarter returns, even after controlling for size, book-to-market, liquidity, S&P 500 membership, momentum, age, and yield.

8.3 Smart institutions versus demand shocks

The positive relation between IO_t and future returns could mean that institutions are smart stock pickers. Alternatively, it could mean that stocks with high institutional ownership are expected to receive larger demand shocks when institutional inflows arrive.

The paper separates IO_t into lagged ownership IO_{t-1} and the change in ownership ΔIO_t . If institutions are smart, changes in holdings should be especially informative. If the demand-shock mechanism is right, lagged ownership should be a useful proxy because stable preferences identify which stocks will benefit from future institutional inflows.

The evidence favors the demand-shock interpretation. Lagged institutional ownership remains significant, while changes in institutional ownership are weaker.

8.4 High-inflow versus low-inflow quarters

A second test uses quarters with high and low aggregate institutional inflows. Institutional inflows are defined as

$$Inflows_t = \sum_i size_{i,t-1} \Delta IO_{i,t},$$

normalized by total market capitalization.

If demand shocks drive the result, the coefficient on institutional ownership should be higher in high-inflow quarters. This is exactly what the paper finds. The coefficient on IO_t is large and significant in high-inflow quarters, but near zero in low-inflow quarters.

This is one of the paper's most important pieces of evidence because it links return predictability to the arrival of institutional capital, not just to the level of institutional ownership.

9 Size premium and demand-shift calibration

9.1 The size-premium reversal

From 1926 to 1979, small stocks earned a large premium over large stocks, about 4 percent per year. From 1980 to 1996, this pattern reversed: large stocks earned higher returns than small stocks.

The paper asks whether the shift toward institutional ownership can account for this reversal. Since institutions prefer large stocks and individuals hold relatively more small stocks, rising institutional ownership should increase relative demand for large stocks.

9.2 Demand decomposition

In March 1980, institutions held 83.1 percent of their portfolios in large stocks and 0.4 percent in small stocks. Individuals held 69.7 percent in large stocks and 3.8 percent in small stocks. Institutions controlled 26.8 percent of the market, while individuals controlled 73.2 percent.

Using a total market value of \$942 billion, demand for large stocks in 1980 is decomposed as

$$D_{Large,1980} = (0.831 \times 0.268 + 0.697 \times 0.732) \times 942 = 690.4.$$

Demand for small stocks in 1980 is

$$D_{Small,1980} = (0.004 \times 0.268 + 0.038 \times 0.732) \times 942 = 27.2.$$

Holding budget shares, prices, and total market value fixed, but raising institutional market share to 51.6 percent, the implied 1996 demands are

$$D_{Large,1996} = (0.831 \times 0.516 + 0.697 \times 0.484) \times 942 = 721.7,$$

$$D_{Small,1996} = (0.004 \times 0.516 + 0.038 \times 0.484) \times 942 = 19.3.$$

Large-stock demand rises by 4.5 percent, while small-stock demand falls by 29.1 percent. The relative demand for large stocks versus small stocks therefore rises by about 47.4 percent.

9.3 Price impact

If the absolute demand elasticity for large and small stocks is ϵ , the relative price impact is

$$\Delta P = \frac{0.474}{\epsilon}.$$

Under unit-elastic demand, the relative price increase of large stocks versus small stocks is 47.4 percent. Spread over the 16.7-year sample period, this equals about 2.3 percent per year:

$$\frac{\log(1.474)}{16.7} \approx 0.023.$$

This is large enough to explain nearly the whole large-stock return advantage over the sample period.

10 Interpretation and limitations

10.1 The paper is about price pressure, not superior institutional skill

The results do not imply that institutional investors earn abnormal profits after fees and trading costs. The analysis is about raw returns on the underlying stocks in institutional portfolios. Institutions may hold stocks that earn high raw returns because their own growing demand pushes up prices, even if fund investors do not earn abnormal net performance.

10.2 Demand curves need not be perfectly elastic

The calibration only works if demand curves for broad groups of stocks are not perfectly elastic. The paper connects to evidence from S&P 500 additions and limits-to-arbitrage models. Arbitrageurs may not fully offset demand shocks because shorting large stocks and buying small stocks involves fundamental risk, liquidity risk, and investor-withdrawal risk.

10.3 Supply responses are limited

One objection is that firms could create more large-stock supply through mergers or equity issuance. The authors argue that supply did not offset the demand shock. In fact, the number of small firms grew dramatically while the number of large firms changed little, and the relative book-value quantity of large stocks to small stocks fell.

10.4 Out-of-sample implication

If rising institutional ownership creates high realized returns for large stocks through price appreciation, then this does not imply large stocks should always have high expected returns. Once institutional ownership stabilizes, the expected premium of large stocks over small stocks should fall. This distinction between in-sample realized returns and future expected returns is essential.

11 Tables and figures

11.1 Table I: Time Series of the Number of Institutions by Manager Type

11.2 Table II: Time Series of the Market Value of Institutional Holdings by Manager Type

TABLE I TIME SERIES OF THE NUMBER OF INSTITUTIONS BY MANAGER TYPE											
Banks		Insurance companies		Mutual fund companies		Investment advisors		Other (university endowments and pensions)		Total	
Number of institutions	Percent of total	Number of institutions	Percent of total	Number of institutions	Percent of total	Number of institutions	Percent of total	Number of institutions	Percent of total	Number of institutions	
Dec-80	216	41.1%	65	12.4%	47	9.0%	122	23.2%	75	14.3%	525
Dec-81	215	38.5%	60	10.8%	51	9.1%	150	26.9%	82	14.7%	558
Dec-82	216	37.4%	60	10.4%	52	9.0%	172	29.8%	78	13.5%	578
Dec-83	226	35.4%	63	9.9%	52	8.1%	218	34.1%	80	12.5%	639
Dec-84	225	32.5%	63	9.1%	51	7.4%	266	38.4%	88	12.7%	693
Dec-85	224	29.2%	69	9.0%	54	7.0%	332	43.3%	87	11.4%	766
Dec-86	208	25.8%	65	8.1%	60	7.4%	379	47.0%	94	11.7%	806
Dec-87	211	24.0%	72	8.2%	58	6.6%	441	50.1%	99	11.2%	881
Dec-88	214	24.3%	62	7.0%	58	6.6%	454	51.6%	92	10.5%	880
Dec-89	218	23.3%	69	7.4%	54	5.8%	506	54.0%	90	9.6%	937
Dec-90	216	22.1%	73	7.5%	57	5.8%	541	55.4%	89	9.1%	976
Dec-91	212	21.1%	70	7.0%	56	5.6%	584	58.1%	83	8.3%	1005
Dec-92	216	19.7%	70	6.4%	63	5.7%	666	60.7%	83	7.6%	1098
Dec-93	191	18.5%	64	6.2%	61	5.9%	649	62.8%	69	6.7%	1034
Dec-94	195	17.2%	75	6.6%	54	4.8%	740	65.2%	71	6.3%	1135
Dec-95	202	15.5%	78	6.0%	96	7.4%	845	65.0%	79	6.1%	1300
Dec-96	172	13.2%	69	5.3%	90	6.9%	900	69.1%	72	5.5%	1303

This table shows the total number of institutions reporting their holdings, broken down by manager type, at the end of each year from 1980 to 1996. Manager types are classified by Spectrum based on the primary business of the institution. The five types are (1) bank, (2) insurance company, (3) investment company (mutual fund), (4) investment adviser, and (5) other. The investment adviser category includes most of the large brokerage firms. The "other" category includes pension funds and university endowments.

TABLE II TIME SERIES OF THE MARKET VALUE OF INSTITUTIONAL HOLDINGS BY MANAGER TYPE													
Banks		Insurance companies		Mutual fund companies		Investment advisors		Other (university endowments and pensions)		Total			
Market value	Percent of total	Market value	Percent of total	Market value	Percent of total	Market value	Percent of total	Market value	Percent of total	Market value	Percent of total	Market value	Percent of total
Dec-80	\$173.00	46.1%	\$ 44.67	11.9%	\$ 30.76	8.2%	\$ 79.90	21.3%	\$ 47.34	12.6%	\$ 375.67	28.4%	
Dec-81	\$146.49	42.3%	\$ 41.64	12.0%	\$ 27.28	7.9%	\$ 89.06	25.7%	\$ 41.82	12.1%	\$ 346.29	28.1%	
Dec-82	\$175.87	40.1%	\$ 52.85	12.0%	\$ 36.05	8.2%	\$ 125.89	28.7%	\$ 48.35	11.0%	\$ 439.01	31.1%	
Dec-83	\$213.39	38.6%	\$ 57.70	10.4%	\$ 47.56	8.6%	\$ 176.43	31.7%	\$ 58.97	10.7%	\$ 553.06	31.8%	
Dec-84	\$215.11	38.1%	\$ 51.33	9.1%	\$ 42.50	7.5%	\$ 180.60	33.6%	\$ 66.25	11.7%	\$ 564.78	33.6%	
Dec-85	\$278.81	36.5%	\$ 64.89	8.5%	\$ 50.82	6.7%	\$ 272.35	35.7%	\$ 96.82	12.7%	\$ 763.49	36.5%	
Dec-86	\$317.32	36.0%	\$ 71.04	8.1%	\$ 50.09	5.7%	\$ 329.84	37.4%	\$ 114.12	12.9%	\$ 882.41	37.5%	
Dec-87	\$298.89	33.8%	\$ 71.24	8.1%	\$ 53.18	6.0%	\$ 346.97	39.2%	\$ 114.31	12.9%	\$ 884.58	38.3%	
Dec-88	\$325.06	32.7%	\$ 79.23	8.0%	\$ 56.55	5.7%	\$ 394.44	39.7%	\$ 138.69	13.9%	\$ 994.27	39.6%	
Dec-89	\$399.28	32.4%	\$ 99.23	8.0%	\$ 69.12	5.6%	\$ 514.40	41.7%	\$ 151.34	12.3%	\$ 1,233.37	40.4%	
Dec-90	\$356.97	30.0%	\$ 92.81	7.8%	\$ 75.86	6.4%	\$ 522.56	43.9%	\$ 143.24	12.0%	\$ 1,191.43	43.3%	
Dec-91	\$481.01	29.6%	\$ 121.44	7.5%	\$ 151.07	9.3%	\$ 693.38	42.6%	\$ 179.75	11.1%	\$ 1,625.65	43.7%	
Dec-92	\$511.16	27.0%	\$ 139.08	7.3%	\$ 202.88	10.7%	\$ 820.85	43.4%	\$ 219.20	11.6%	\$ 1,893.36	46.0%	
Dec-93	\$507.96	25.4%	\$ 180.54	9.0%	\$ 255.00	12.8%	\$ 853.47	42.7%	\$ 201.48	10.1%	\$ 1,998.45	42.8%	
Dec-94	\$557.89	25.1%	\$ 215.26	9.7%	\$ 292.31	13.2%	\$ 946.72	42.6%	\$ 207.61	9.4%	\$ 2,219.78	48.0%	
Dec-95	\$727.79	22.4%	\$ 324.50	10.0%	\$ 719.53	22.2%	\$ 1,174.94	36.2%	\$ 295.97	9.1%	\$ 3,242.62	51.2%	
Dec-96	\$860.93	21.0%	\$ 373.79	9.4%	\$ 1,008.87	25.3%	\$ 1,479.37	37.2%	\$ 259.93	6.5%	\$ 3,981.89	51.6%	

This table shows the total market value of equity held by each manager type at the end of each year from 1980 to 1996. Manager types are classified by Spectrum based on the primary business of the institution. The five types are (1) bank, (2) insurance company, (3) investment company (mutual fund), (4) investment adviser, and (5) other. The investment adviser category includes most of the large brokerage firms. The "other" category includes pension funds and university endowments. All figures are in billions of nominal dollars.

Table I: Time Series of the Number of Institutions by Manager Type

Table II: Time Series of the Market Value of Institutional Holdings by Manager Type

Read for Table I: Table I documents the institutional-sector composition by number of managers. The total number of 13F institutions rises from 525 in 1980 to 1,303 in 1996. The most important shift is the rise of investment advisers: they increase from 122 institutions, or 23.2 percent of the total, to 900 institutions, or 69.1 percent. Banks move in the opposite direction, falling from 41.1 percent to 13.2 percent of reporting institutions.

Read for Table II: Table II shows that the institutional share of the equity market nearly doubles, from 26.8 percent in March 1980 to 51.6 percent in December 1996. In dollar terms, holdings rise from hundreds of billions to nearly \$4 trillion. The table also shows that mutual funds and investment advisers become much more important owners over time, while banks lose relative share.

11.3 Figure I: Percent of Market Owned by All Institutions and by the First, First Ten, and First One Hundred as Ranked by Holdings

11.4 Table III: Cross-Sectional Correlations

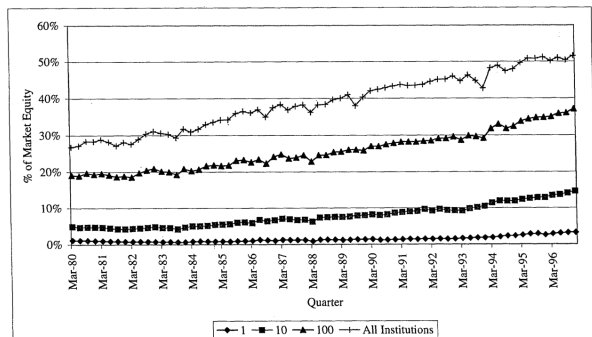


FIGURE I

Percent of Market Owned by All Institutions and by the First, First Ten, and First One Hundred as Ranked by Holdings

The figure plots the cumulative institutional holdings (as a fraction of the market value of all equities) for all institutions, and for the first, first ten, and first one hundred institutions as ranked by total holdings.

TABLE III
CROSS-SECTIONAL CORRELATIONS

	Institutional ownership	Book-to-market	Market equity	Volatility	Turnover	Price	S&P 500	Momentum _{-3,0}	Momentum _{-12,-3}	Age	Yield
Institutional ownership	1.000 [0.000]										
Book-to-market	0.026 [0.007]	1.000 [0.000]									
Size	0.625 [0.003]	-0.063 [0.011]	1.000 [0.000]								
Volatility	-0.309 [0.003]	-0.256 [0.012]	-0.476 [0.007]	1.000 [0.000]							
Turnover	0.038 [0.011]	-0.202 [0.008]	0.240 [0.012]	0.223 [0.012]	1.000 [0.000]						
Price	0.532 [0.003]	0.064 [0.012]	0.785 [0.003]	-0.697 [0.010]	0.135 [0.013]	1.000 [0.000]					
S&P 500	0.388 [0.004]	0.032 [0.007]	0.542 [0.001]	-0.238 [0.004]	0.117 [0.007]	0.315 [0.002]	1.000 [0.000]				
Momentum _{-3,0}	0.071 [0.008]	0.057 [0.017]	0.172 [0.011]	0.017 [0.024]	0.147 [0.015]	0.256 [0.014]	0.014 [0.008]	1.000 [0.000]			
Momentum _{-12,-3}	0.035 [0.010]	0.064 [0.012]	0.118 [0.014]	0.020 [0.023]	-0.020 [0.011]	0.157 [0.017]	0.013 [0.006]	0.028 [0.010]	1.000 [0.000]		
Age	0.238 [0.003]	0.314 [0.010]	0.241 [0.006]	-0.296 [0.008]	-0.088 [0.010]	0.206 [0.012]	0.215 [0.005]	0.031 [0.010]	0.025 [0.009]	1.000 [0.000]	
Yield	0.284 [0.009]	0.290 [0.012]	0.457 [0.009]	-0.609 [0.005]	-0.123 [0.014]	0.462 [0.007]	0.297 [0.003]	0.008 [0.017]	0.038 [0.015]	0.353 [0.006]	1.000 [0.000]

The table shows the average cross-sectional correlation between institutional ownership and firm characteristics and for all pairs of these characteristics. The average is taken over the 60 quarters starting with the first quarter in 1980 through the last quarter of 1996. Standard errors for the correlation coefficients are in brackets. All variables except institutional ownership, S&P 500 membership, and momentum are expressed in natural logarithms. The book-to-market ratio is the book value for the fiscal year ended before the most recent June 30, divided by size as of December 31 during that fiscal year. Size is quarter-end market capitalization. Volatility is the variance of monthly returns over the previous two years. Turnover is volume divided by shares outstanding, measured for the month prior to the beginning of the quarter. Price is the quarter-end share price. S&P 500 is a dummy variable equal to one if the firm is included in the S&P 500 at the quarter-end, and zero otherwise. Momentum_{-3,0} is the firm's past three-month gross return. Momentum_{-12,-3} is the firm's past nine-month gross return. Yield is the yield on the firm's debt.

Figure I: Percent of Market Owned by All Institutions and by the First, First Ten, and First One Hundred as Ranked by Holdings

Table III: Cross-Sectional Correlations

Read for Figure I: Figure I shows that institutional ownership rises steadily and that the increase is concentrated among the biggest managers. The share controlled by all institutions rises above 50 percent, while the top 100 institutions alone reach roughly 37 percent by 1996. This concentration is important because the price effect comes from large coordinated shifts in investor clientele.

Read for Table III: Table III shows the raw correlations among institutional ownership and firm characteristics. Institutional ownership is strongly positively correlated with size, price, S&P 500 membership, turnover, and age. Because many characteristics are also correlated with size, the table motivates the multivariate regressions in Table IV.

11.5 Table IV: Determinants of Institutional Ownership

11.6 Figure II: Difference in Return on Institutional Portfolio Relative to Individuals' Portfolio

TABLE IV
DETERMINANTS OF INSTITUTIONAL OWNERSHIP

Variable	Average coefficient	Number positive [significant]	Number negative [significant]
Constant	-0.4478	3 [0]	65 [62]
Book-to-market	0.0109	52 [46]	16 [10]
Size	0.0403	68 [68]	0 [0]
Volatility	0.0040	51 [31]	17 [9]
Turnover	0.0002	59 [51]	9 [3]
Price	0.0469	68 [68]	0 [0]
S&P 500	0.0537	61 [58]	7 [3]
Momentum _{-3,0}	-0.0358	0 [0]	68 [64]
Momentum _{-12,-3}	-0.0486	2 [0]	66 [58]
Age	0.0124	65 [51]	3 [1]
Yield	-0.0192	2 [0]	66 [43]

The table summarizes the results from 88 quarterly (cross-sectional) regressions for the sample period. The dependent variable is the institutional ownership as a percentage of the firm's market capitalization (IO). All variables except institutional ownership, S&P 500 membership, and momentum are expressed in natural logarithms. The book-to-market ratio is the book value for the fiscal year ended before the most recent June 30, divided by size as of December 31 during that fiscal year. Size is quarter-end market capitalization. Volatility is the variance of monthly returns over the previous two years. Turnover is volume divided by shares outstanding, measured for the month prior to the beginning of the quarter. Price is the quarter-end share price. S&P 500 is a dummy variable equal to one if the firm is included in the S&P 500 at the quarter-end, and zero otherwise. Momentum_{-3,0} is the firm's past three-month gross return. Momentum_{-12,-3} is the nine-month gross return prior to the beginning of the filing quarter. Age is the number of months since the first return for this stock appears in CRSP file. Yield is the cash dividends for the fiscal year ended before the most recent June 30, divided by size as of December 31 in that fiscal year. The table gives the number of positive coefficients, number of negative coefficients, the number of significantly positive coefficients (at the 95 percent confidence level) and the number of significantly negative coefficients (at the 95 percent confidence level). Significance of the quarterly coefficients is computed using White-corrected standard errors (1980). The sample also ranges from 1979 to March 1980 to February 1986.

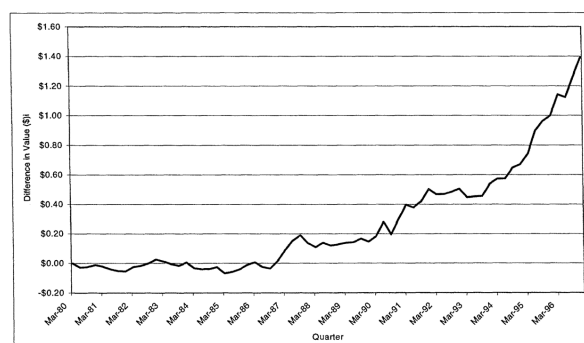


FIGURE II

Difference in Return on Institutional Portfolio Relative to Individuals' Portfolio

The figure tracks the difference in the value of one dollar invested on March 31, 1980, in the portfolio of stocks held by institutions versus an equivalent investment in the portfolio of stocks held by individuals. Quarterly returns are calculated as the value-weighted return on all stocks held by institutions (or individuals) at the beginning of a quarter.

Table IV: Determinants of Institutional Ownership

Figure II: Difference in Return on Institutional Portfolio Relative to Individuals' Portfolio

Read for Table IV: Table IV is the main ownership-preference table. Institutional ownership is consistently higher for large stocks, high-price stocks, high-turnover stocks, S&P 500 stocks, and older firms. The two momentum coefficients are negative in almost every quarter, so institutions are not momentum investors. Overall, institutions prefer large, liquid, established stocks with low past returns.

Read for Figure II: Figure II plots the cumulative dollar difference between the aggregate institutional portfolio and the aggregate individual portfolio. The institutional portfolio pulls ahead over the sample period, especially after the late 1980s. This figure motivates the return regressions by showing that stocks held by institutions outperform stocks held by individuals in raw value-weighted terms.

11.7 Table V: Institutional Ownership and Returns

TABLE V
INSTITUTIONAL OWNERSHIP AND RETURNS

Number of cross-sectional regressions	<i>Full sample</i>	<i>Full sample</i>	<i>High inflow sample</i>	<i>Low inflow sample</i>
	Average coefficient [time series <i>t</i> -statistic]			
	67	66	33	34
Intercept	1.0439 [−19.11]	1.0458 [19.03]	1.0060 [28.82]	1.0806 [15.69]
Book-to-market	0.0017 [0.51]	0.0020 [0.62]	0.0006 [0.17]	0.0027 [0.86]
Size	−0.0031 [−2.02]	−0.0029 [−1.89]	−0.0009 [−0.59]	−0.0053 [−3.32]
Volatility	−0.0072 [−1.69]	−0.0072 [−1.68]	−0.0026 [−0.71]	−0.0117 [−2.44]
Turnover	−0.0033 [−1.99]	−0.0032 [−1.89]	−0.0022 [−1.28]	−0.0044 [−2.73]
Price	0.0009 [0.37]	0.0010 [0.41]	0.0006 [0.26]	0.0012 [0.45]
S&P 500	0.0051 [1.93]	0.0057 [2.16]	0.0053 [2.18]	0.0049 [1.70]
Momentum _{−3,0}	−0.0071 [−0.51]	−0.0079 [−0.57]	0.0053 [0.40]	−0.0191 [−1.34]
Momentum _{−12,−3}	0.0337 [4.22]	0.0327 [4.12]	0.0285 [3.60]	0.0388 [4.77]
Age	−0.0084 [−0.87]	−0.0106 [−1.13]	−0.0021 [−0.85]	−0.0146 [−1.08]
Yield	−0.0005 [−0.17]	−0.0008 [−0.25]	−0.0046 [−1.39]	0.0034 [1.18]
Institutional ownership (IO_t)	0.0116 [2.13]		0.0230 [4.65]	0.0006 [0.10]
Last period's institutional ownership (IO_{t-1})		0.0135 [2.49]		
Change in institutional ownership (ΔIO_t)		0.0385 [1.79]		

This table summarizes the results of cross-sectional regressions of quarterly returns on institutional ownership and other firm characteristics. Each cross-sectional regression is estimated by weighted-least-squares, with weights equal to the prior market value of each stock. The table reports average coefficients, with *t*-statistics for these averages based on their time-series standard deviations. All variables except return, institutional ownership, S&P 500 membership, and momentum are expressed in natural logarithms. The book-to-market ratio is the book value for the fiscal year ended before the most recent June 30, divided by size as of December 31 during that fiscal year. Size is quarter-end market capitalization. Volatility is the variance of monthly returns over the previous two years. Turnover is volume divided by shares outstanding, measured for the month prior to the beginning of the quarter. Price is the quarter-end share price. S&P 500

Table V: Institutional Ownership and Returns

Read for Table V: Table V shows that institutional ownership forecasts returns. In the full sample, the coefficient on institutional ownership is positive and statistically significant. When IO_t is split into lagged ownership and changes in ownership, lagged ownership is more robust, which supports the demand-shock story over the smart-institution story. The high- versus low-inflow split is even more revealing: the ownership coefficient is large and significant only in high-inflow quarters.

12 Key takeaways

12.1 Core conclusion

Institutional ownership is not passive. Because institutions have stable preferences for large and liquid stocks, the growth of institutional ownership changes aggregate demand across stock characteristics and can move relative prices.

12.2 Main empirical finding

Institutions nearly doubled their share of the equity market from 1980 to 1996, and their holdings became increasingly concentrated among the largest managers. Institutional ownership is higher for large, liquid, established stocks and lower for recent winners after controls.

12.3 Asset-pricing implication

The level of institutional ownership predicts future stock returns, especially when institutional inflows are high. This pattern is more consistent with demand pressure than with superior institutional stock-picking skill.

12.4 Size-premium implication

The compositional shift toward institutions raises relative demand for large stocks and reduces demand for small stocks. Under unit-elastic demand, the estimated relative price effect is about 47 percent, or about 2.3 percent per year over the sample period, enough to explain much of the post-1980 reversal of the small-stock premium.

12.5 Bottom line

Investor clienteles can matter for asset prices. When a large and growing investor class prefers certain stock characteristics, factor returns may partly reflect demand pressure rather than compensation for fundamental risk.